

The time-dependent quantum harmonic oscillator: a pedagogical approach via the Lewis-Riesenfeld dynamical invariant method

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In quantum mechanics courses, students often solve the Schrödinger equation for the harmonic oscillator with time-independent parameters. However, time-dependent quantum harmonic oscillators are relevant in modeling several problems as, for instance, the description of quantum motion of particles in traps, shortcuts to adiabaticity, generation of squeezed states, as well as quantum scalar fields evolving in expanding universes. In the present paper, we discuss, with a pedagogical approach, the quantum harmonic oscillator with time-dependent frequency via the Lewis-Riesenfeld dynamical invariant method, revisiting the main steps to obtain the wave function associated with this model, and briefly discussing the relation between this oscillator and the generation of squeezed states. As examples of didactic applications of time-dependent harmonic oscillators and the Lewis-Riesenfeld method in quantum mechanics courses, we solve the following problems: the calculation of the transition probability associated with a harmonic oscillator which undergoes jumps in its frequency, and the analysis of the dynamics of a quantum particle in a Paul trap.

I. INTRODUCTION

The harmonic oscillator is one of the fundamental systems in physics, permeating from classical mechanics [1–4] to the quantum field theory [5, 6]. In the context of undergraduate courses, the harmonic oscillator with time-independent parameters is vastly addressed in books in classical (e.g., in Refs. [1–3]) and in quantum mechanics (e.g., in Refs. [7–9]). Regarding the harmonic oscillator with time-dependent parameters, here denominated as time-dependent harmonic oscillator (TDHO), although it is addressed in relatively few books of classical mechanics (e.g., in Refs. [1, 4]), it is still commonly discussed in classrooms [10]. An example of a system of this nature is found in Ref. [11] (considering a motion with small oscillations), in which there is the illustrative situation of a child playing on a swing which, by bending its knees, periodically modulates its center of mass, resulting in amplification of the amplitude of the motion. On the other hand, the TDHO is more rarely covered in quantum mechanics textbooks (e.g., Refs. [9, 12] include a problem involving the calculation of the transition probability of an oscillator that experiences a sudden jump in its frequency).

TDHOs are important in quantum physics [13–18]. For example, TDHOs appear in the description of the interaction between a spinless charged quantum particle and a time-dependent external classical electromagnetic field [19–21], as well as in the quantization of the free electromagnetic field in nonstationary media [22–26]. In atomic physics, the confinement of charged particles by oscillating electromagnetic fields can create effective potentials of the TDHO type [27–33]. This device, known as a Paul trap, has many applications in modern physics [34]. In this sense, TDHOs can also be used to study the quantum motion of particles in traps. In the framework of quantum field theory in curved spacetimes, the modes of a scalar field, in an expanding universe described by the Friedmann-Robertson-Walker metric, behave like TDHOs whose frequency changes in time due to the evolution of spacetime (see Ref. [35] and Refs. therein). In quantum control techniques and shortcuts to adiabaticity, the oscillator frequency is deliberately manipulated to obtain the desired evolution of the system in finite time, without generating unwanted transitions [36–48]. In the context of quantum optics, squeezed states, used to overcome classical measurement limits (for example, in interferometry), are generated by manipulating systems modelled as TDHOs [49–56]. Therefore it is important to improve discussions on TDHOs in the context of undergraduate quantum mechanics courses, an effort that has already begun in some articles in physics teaching journals, such as those already mentioned [10, 18, 57], and also in Refs. [58, 59].

While the energy is a constant of motion for the harmonic oscillator with time-independent parameters, it is not for the TDHO. However, Lewis [60, 61] found a certain constant of motion for the TDHO, which is nowadays known as the Lewis' dynamical invariant. Lutzky, in Ref. [62], showed how to obtain such Lewis' invariant from Noether's theorem. Considering that Lutzky's work was mostly known by specialists, but not by teachers, and also that the TDHO is an important issue in physics education, Abe *et al.* [10] reviewed Lutzky's work in a didactic manner, interconnecting, in a simple way, the TDHO, the concept of invariant, and Noether's theorem.

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In the context of quantum mechanics, invariant operators, which can be viewed as generalizations of constants of motion, are useful for finding exact solutions of the Schrödinger equation for systems whose Hamiltonians explicitly depend on time [57]. As an example, one can cite the method proposed by Lewis and Riesenfeld [14], which uses invariant operators to solve the Schrödinger equation for generic time-dependent Hamiltonians and has already been applied in several problems in physics, including those which can be described by a Hamiltonian of a TDHO [14, 22, 37, 63–72]. We can also cite the method discussed by Andrews in Ref. [57], which obtained invariant operators similar to those used by Lewis and Riesenfeld [14], and, with a didactic approach, used them to solve the Schrödinger equation for time-dependent Hamiltonians that are quadratic in position and momentum, with the case of a Hamiltonian describing a TDHO being a particular application.

There are also other methods, apart from the LR method, that make it possible to obtain exact solutions of the Schrödinger equation for TDHOs. For example, Ref. [73] analysed a TDHO with a general time-dependent quadratic Hamiltonian (which can represent both a TDHO with time-dependent mass and frequency, and one with only time-dependent frequency) via the path-integral method. Something similar is done in Ref. [74]. In Ref. [27], the wave function associated with a TDHO with time-dependent frequency was found using canonical transformations and, in Ref. [75], was obtained using the Madelung-Bohm approach to quantum mechanics. In Refs. [76–78], the TDHO with time-dependent frequency was studied using algebraic methods. In this way, studying the TDHO also offers the opportunity to discuss important concepts, such as invariants operators, path-integral and algebraic methods, among others, which are increasingly relevant elements in contemporary physics, both theoretical and experimental.

In the present paper, aiming to contribute to the improvement of the discussions on TDHOs, especially in quantum mechanics courses, we didactically discuss the Lewis-Riesenfeld (LR) method [14, 60, 61] and its application to the case of a harmonic oscillator with time-dependent frequency, describing the step-by-step use of the method. As part of our didactic strategy, we map these steps into those already used by undergraduate students to solve the Schrödinger equation for the time-independent harmonic oscillator. As particular applications, we discuss the calculation of the transition probability associated with a TDHO which undergoes a sudden jump in its frequency (the mentioned problem found in Refs. [9, 12]), and we briefly discuss the dynamics of a quantum particle in a Paul trap, which is also modeled by a TDHO with a time-dependent frequency (from the classical point of view, the problem of particles in a Paul trap has been investigated in several physics teaching articles [79–87]). These calculations can be used as didactic examples of applications of the TDHOs and the LR method in quantum mechanics courses.

This paper is organized as follows. In Sec. II, to prepare the reader for the subsequent sections, we briefly review the procedures for obtaining the wave function associated with the quantum harmonic oscillator with time-independent parameters, as is usually done in quantum mechanics textbooks. In Sec. III, we provide a detailed pedagogical review of the LR method for a generic Hamiltonian $\hat{H}(t)$. In Sec. IV, we provide a review of the application of the LR method to the case of a TDHO with a generic time-dependent frequency. More specifically: In Sec. IV A, we review the calculation of the wave function (via the LR method) associated with this model. In Sec. IV B, we discuss the generation of squeezed states by a TDHO with time-dependent frequency. In Sec. IV C, we use these concepts to solve the problem of the calculation of the transition probability associated to a quantum harmonic oscillator that undergoes a sudden jump in its frequency. In Sec. IV D, we also use all these tools to discuss the problem of the dynamics of a quantum particle in a Paul trap. In Sec. V, we present our final remarks.

II. A BRIEF REVIEW OF THE TIME-INDEPENDENT QUANTUM HARMONIC OSCILLATOR

As will be seen later, although the solution of the time-dependent quantum harmonic oscillator needs a more robust method to lead with, it has several similarities with the time-independent correspondent problem. In this way, we start by reviewing the problem of the time-independent quantum harmonic oscillator, which is a system that is widely studied in quantum mechanics courses [7, 9].

First, we consider the Schrödinger equation for the one-dimensional time-independent harmonic oscillator,

$$i\hbar \frac{\partial}{\partial t} |\psi^{(0)}(t)\rangle = \hat{H}_0 |\psi^{(0)}(t)\rangle, \quad (1)$$

with the Hamiltonian operator defined by

$$\hat{H}_0 = \frac{\hat{p}^2}{2m_0} + \frac{1}{2}m_0\omega_0^2\hat{x}^2, \quad (2)$$

where m_0 and ω_0 are the time-independent mass and angular frequency, respectively. The position and momentum operators, given by $\hat{x}$ and $\hat{p} = -i\hbar\partial/\partial x$ (in position space), satisfy the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ [7, 9]. Besides this, it is straight to note that, for the case of a time-independent harmonic oscillator, we have that $\hat{H}_0$ is an invariant operator, i.e., $d\hat{H}_0/dt = 0$.

The solution of Eq. (1) in the position space, given by $\Psi^{(0)}(x, t) = \langle x | \psi^{(0)}(t) \rangle$, can be written as [7, 9]

$$\Psi^{(0)}(x, t) = \sum_{n=0}^{\infty} C_n^{(0)} \Psi_n^{(0)}(x, t), \quad (3)$$

where $\Psi_n^{(0)}(x, t)$ are the wave functions associated to each allowed value of energy (E_n) of the system, and the coefficients $C_n^{(0)}$ depends only on the initial conditions. Using this solution into Eq. (1) in the position space, and performing the method of separation of variables, one finds (see Appendix A for more details) [7, 9]

$$\Psi_n^{(0)}(x, t) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m_0 \omega_0}{\pi \hbar} \right)^{\frac{1}{4}} \exp \left[-i \left(n + \frac{1}{2} \right) \omega_0 t \right] \exp \left(-\frac{m_0 \omega_0 x^2}{2 \hbar} \right) \mathcal{H}_n \left(\sqrt{\frac{m_0 \omega_0}{\hbar}} x \right). \quad (4)$$

We highlight that, for a TDHO [where $\hat{H}_0 \rightarrow \hat{H}(t)$], the LR method prescribes that the wave function can be found by means of a sequence of steps similar to these ones used in the case of a time-independent harmonic oscillator (see Appendix A). In the time-independent case, $\hat{H}_0$ is an invariant operator, since $d\hat{H}_0/dt = 0$. In the time-dependent case, the Hamiltonian is not an invariant operator anymore, but, according to the LR method, we can find another operator, denoted by $\hat{I}(t)$, so that this new operator is invariant, i.e., $d\hat{I}(t)/dt = 0$. Thus, by finding $\hat{I}(t)$, the next steps are similar to those shown in Appendix A, as discussed next.

III. LEWIS-RIESENFELD DYNAMICAL INVARIANT METHOD IN QUANTUM MECHANICS

Before discussing the solution of a TDHO, let us start reviewing the well known LR dynamical invariant method [14, 60, 61], which consists of a mechanism for finding exact solutions of the Schrödinger equation for systems whose Hamiltonians explicitly depend on time, i.e., for finding solutions of the following equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle. \quad (5)$$

The starting point for solving this equation consists in finding a Hermitian operator $\hat{I}(t)$, called invariant operator. Such operator has the property of being a constant of motion, i.e., $d\hat{I}(t)/dt = 0$, such that, from the Heisenberg equation [7, 14], one has

$$\frac{\partial \hat{I}(t)}{\partial t} + \frac{1}{i\hbar} [\hat{I}(t), \hat{H}(t)] = 0. \quad (6)$$

According to Ref. [14]: “The action of the invariant operator on a Schrödinger state vector produces another solution of the Schrödinger equation. This result is valid for any invariant, even if the latter involves the operation of time differentiation”. To see this, one has to apply the operator $\partial \hat{I}(t)/\partial t$ on the state $|\psi(t)\rangle$, so that, by using Eq. (6), one obtains

$$\frac{\partial \hat{I}(t)}{\partial t} |\psi(t)\rangle = -\frac{1}{i\hbar} [\hat{I}(t) \hat{H}(t) - \hat{H}(t) \hat{I}(t)] |\psi(t)\rangle. \quad (7)$$

Using Eq. (5) in this equation, one sees that

$$i\hbar \frac{\partial}{\partial t} [\hat{I}(t) |\psi(t)\rangle] = \hat{H}(t) [\hat{I}(t) |\psi(t)\rangle], \quad (8)$$

which shows that $\hat{I}(t) |\psi(t)\rangle$ is also a solution of the Schrödinger equation.

Following Ref. [14]: “We assume that the invariant operator is one of a complete set of commuting observables, so that there is a complete set of eigenstates of $\hat{I}(t)$ ”. In this way, let us denote the eigenvalues of $\hat{I}(t)$ as λ , which are time-independent (see Appendix B for more details) [14], and its orthonormal eigenstates associated to λ as $|\lambda, \zeta; t\rangle$, with ζ being all quantum numbers, other than λ , needed to specify the eigenstates. Therefore, one writes (for a discrete spectrum)

$$\hat{I}(t) |\lambda, \zeta; t\rangle = \lambda |\lambda, \zeta; t\rangle, \quad (9)$$

$$\sum_{\lambda, \zeta} |\lambda, \zeta; t\rangle \langle \lambda, \zeta; t| = 1, \quad (10)$$

$$\langle \lambda', \zeta'; t | \lambda, \zeta; t \rangle = \delta_{\lambda', \lambda} \delta_{\zeta', \zeta}, \quad (11)$$

where $\delta_{i,j}$ is the Kronecker delta (for a continuous spectrum, this is replaced by Dirac delta functions).

We remark that the eigenstates $|\lambda, \zeta; t\rangle$ do not satisfy, in general, the Schrödinger equation, i.e.,

$$i\hbar \frac{\partial}{\partial t} |\lambda, \zeta; t\rangle \neq \hat{H}(t) |\lambda, \zeta; t\rangle. \quad (12)$$

Nevertheless, each state $|\lambda, \zeta; t\rangle$ can be multiplied by a phase factor, resulting in a state $|\lambda, \zeta; t\rangle_\alpha$, such that

$$|\lambda, \zeta; t\rangle_\alpha = \exp[i\alpha_{\lambda,\zeta}(t)] |\lambda, \zeta; t\rangle, \quad (13)$$

with $\alpha_{\lambda,\zeta}(t)$ being a real and time-dependent phase function. Furthermore, following Ref. [14]: “If the invariant does not involve time differentiation, then we are able to derive a simple and explicit rule for choosing the phases of the eigenstates of $\hat{I}(t)$ such that these states themselves satisfy the Schrödinger equation”. In this way, assuming that the invariant operator does not involve time differentiation [14], it is straightforward to see that the states $|\lambda, \zeta; t\rangle_\alpha$ are also eigenstates of $\hat{I}(t)$, and form an orthonormal basis. Besides this, if we require that the state vectors $|\lambda, \zeta; t\rangle_\alpha$ are solutions of the Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} |\lambda, \zeta; t\rangle_\alpha = \hat{H}(t) |\lambda, \zeta; t\rangle_\alpha, \quad (14)$$

then, one obtains that the phase functions $\alpha_{\lambda,\zeta}(t)$ have to obey the following equation:

$$\frac{d\alpha_{\lambda,\zeta}(t)}{dt} = \langle \lambda, \zeta; t | \left[i \frac{\partial}{\partial t} - \frac{1}{\hbar} \hat{H}(t) \right] | \lambda, \zeta; t \rangle. \quad (15)$$

The solution $\alpha_{\lambda,\zeta}(t)$ of the equation above is known as the Lewis-Riesenfeld phase and, in turn, is composed of the dynamical and geometric phases. These phases have applications in different areas of physics (see, for example, Refs. [88–91]).

The original problem of finding the eigenstates and eigenvalues of $\hat{H}(t)$ is replaced by the problem of finding the eigenstates and eigenvalues of $\hat{I}(t)$ and the phase functions $\alpha_{\lambda,\zeta}(t)$. By knowing the operator $\hat{I}(t)$ (which is, in general, obtained from an ansatz), one can calculate its eigenvalues and eigenstates by means of Eq. (9), and, with this, obtain the phase functions $\alpha_{\lambda,\zeta}(t)$ from Eq. (15). Therefore, the general solution $|\psi(t)\rangle$ of Eq. (5) can be described as a linear combination of $|\lambda, \zeta; t\rangle$ as

$$|\psi(t)\rangle = \sum_{\lambda,\zeta} C_{\lambda,\zeta} \exp[i\alpha_{\lambda,\zeta}(t)] |\lambda, \zeta; t\rangle, \quad (16)$$

in which the time-independent coefficients $C_{\lambda,\zeta}$ depend on the initial conditions. For more details on the steps between Eqs. (13)-(16), see Appendix C.

In summary, the LR method consists of the following steps:

1. One proposes a Hermitian invariant operator $\hat{I}(t)$, which satisfies Eq. (6), and does not contain time derivatives;
2. From Eq. (9), one calculates the eigenvalues λ and eigenstates $|\lambda, \zeta; t\rangle$ associated to $\hat{I}(t)$;
3. Using Eq. (15), one obtains the phase functions $\alpha_{\lambda,\zeta}(t)$;
4. Finally, by means of Eq. (16), one calculates the state vector $|\psi(t)\rangle$.

As a consistency check of Eq. (16), let us discuss the situation in which we have a time-independent Hamiltonian [$\hat{H}(t) = \hat{H}_{\text{TI}}$]. In this situation, one can propose that the invariant operator is given by $\hat{I}(t) = \hat{I}_{\text{TI}}$, with $\hat{I}_{\text{TI}}$ being a time-independent operator. This results in $\partial \hat{I}(t) / \partial t = \partial \hat{I}_{\text{TI}} / \partial t = 0$, which, from Eq. (6), leads to $[\hat{I}(t), \hat{H}(t)] = [\hat{I}_{\text{TI}}, \hat{H}_{\text{TI}}] = 0$. As a consequence, the eigenstates of $\hat{H}_{\text{TI}}$, denoted by $|E_j\rangle$, are also eigenstates of $\hat{I}_{\text{TI}}$ [7], which means that the state $|\lambda, \zeta; t\rangle$ coincides with $|E_j\rangle$, so that we can do the mapping $|\lambda, \zeta; t\rangle \rightarrow |E_j\rangle$ in Eq. (16) and write

$$|\psi(t)\rangle = \sum_j C_j \exp[i\alpha_j(t)] |E_j\rangle. \quad (17)$$

The same mapping has to be made in Eq. (15) in order to obtain the phase function for this case. Since the state $|E_j\rangle$ is time-independent [$\partial |E_j\rangle / \partial t = 0$] and using the eigenvalue equation $\hat{H}_{\text{TI}} |E_j\rangle = E_j |E_j\rangle$, where E_j are the eigenvalues of $\hat{H}_{\text{TI}}$, one can find the solution of Eq. (15), which is given by

$$\alpha_j(t) = -\frac{E_j t}{\hbar}. \quad (18)$$

By substituting this result into Eq. (17), one sees that we recover the general result for $|\psi(t)\rangle$ commonly presented in quantum mechanics textbooks [7, 9] in the context of problems involving time-independent Hamiltonians.

By comparing the characteristics of a time-dependent problem within the context of the LR method with the corresponding time-independent case, we can find correlations, which are summarized in Table I.

	Time-independent case	Time-dependent case
Schrödinger equation	$i\hbar\partial_t \psi(t)\rangle = \hat{H}_{\text{TI}} \psi(t)\rangle$	$i\hbar\partial_t \psi(t)\rangle = \hat{H}(t) \psi(t)\rangle$
Invariant operator	$\hat{H}_{\text{TI}}$	$\hat{I}(t)$
Eigenvalue equation	$\hat{H}_{\text{TI}} E_j\rangle = E_j E_j\rangle$	$\hat{I}(t) \lambda, \zeta; t\rangle = \lambda \lambda, \zeta; t\rangle$
General solution	$ \psi(t)\rangle = \sum_j C_j \exp[i\alpha_j(t)] E_j\rangle$	$ \psi(t)\rangle = \sum_{\lambda, \zeta} C_{\lambda, \zeta} \exp[i\alpha_{\lambda, \zeta}(t)] \lambda, \zeta; t\rangle$
Phase function	$\alpha_j(t) = -E_j t/\hbar$	$\alpha_{\lambda, \zeta}(t) = \int_0^t dt' \langle \lambda, \zeta; t' [i\partial_{t'} - \hat{H}(t')/\hbar] \lambda, \zeta; t' \rangle$

Table I. Comparison between the time-independent and time-dependent cases.

IV. THE SOLUTION OF A QUANTUM HARMONIC OSCILLATOR WITH TIME-DEPENDENT FREQUENCY VIA LEWIS-RIESENFELD DYNAMICAL INVARIANT METHOD

In the present section, we analyze the one-dimensional TDHO with time-dependent frequency $\omega(t)$, which is described by the Hamiltonian operator

$$\hat{H}(t) = \frac{\hat{p}^2}{2m_0} + \frac{1}{2}m_0\omega(t)^2\hat{x}^2. \quad (19)$$

For didactic reasons, in this paper we focus only on the case of a time-dependent frequency, but, in the literature, one finds more general models that have been studied. For example, in Refs. [15, 16], one finds the discussion of the case where the frequency and the mass depend on time. In Refs. [92–94], one finds the discussion of the driven generalized TDHO, where there are time-dependent parameters coupled to linear terms in $\hat{x}$ and $\hat{p}$, as well as external forces, in the Hamiltonian operator. In Ref. [95], one finds the discussion of the case where there are dependences of the type $1/\hat{x}$ and $1/\hat{x}^2$, multiplied by time-dependent factors, in the Hamiltonian operator. Finally, in Refs. [96–98], there are cases where there are damped harmonic oscillators in a time-dependent noncommutative space, and, in Ref. [99], there is the case of systems with two coupled TDHOs.

A. Calculation of the wave function

1. The invariant operator

Lewis and Reisenfeld [14] assumed the existence, for a TDHO, of a Hermitian invariant operator $\hat{I}(t)$ of the quadratic form [similar to $\hat{H}(t)$] given by

$$\hat{I}(t) = \frac{1}{2}[\eta_1(t)\hat{x}^2 + \eta_2(t)\hat{p}^2 + \eta_3(t)\{\hat{x}, \hat{p}\}], \quad (20)$$

where $\eta_1(t)$, $\eta_2(t)$ and $\eta_3(t)$ are real time-dependent functions [which ensures that $\hat{I}(t)$ is Hermitian], and $\{\hat{x}, \hat{p}\}$ is the anti-commutator between the operators $\hat{x}$ and $\hat{p}$. (Note that, in the particular case where the mass and the frequency are independent of time, the invariant operator becomes $\hat{I}(t) \rightarrow \hat{I}_0 = \hat{H}_0/\omega_0 = \frac{1}{\omega_0} \left[\frac{1}{2}m_0\omega_0^2\hat{x}^2 + \frac{1}{2m_0}\hat{p}^2 \right]$ [14].) Since the invariant operator must be a constant of motion [$d\hat{I}(t)/dt = 0$], then we require that this ansatz satisfies Eq. (6). Thus, one finds that the invariant operator, associated to a TDHO with time-dependent frequency, is given by [14–16, 60, 61] (for more details, see Appendix D)

$$\hat{I}(t) = \frac{1}{2} \left\{ \left[\frac{\hat{x}}{\rho(t)} \right]^2 + [\rho(t)\hat{p} - m_0\dot{\rho}(t)\hat{x}]^2 \right\}, \quad (21)$$

where the parameter $\rho(t)$ satisfies the so-called Ermakov-Pinney equation [100–103]:

$$\ddot{\rho}(t) + \omega(t)^2\rho(t) = \frac{1}{m_0^2\rho(t)^3}. \quad (22)$$

In this way, by knowing how the frequency of the oscillator depend on time, one can obtain an expression for the invariant operator by only calculating the parameter $\rho(t)$ using Eq. (22). We highlight that although Eq. (22) is a nonlinear differential equation, it has a general exact solution (see Appendix E).

2. The eigenvalues and eigenfunctions of the invariant operator

Following Ref. [14]: “The eigenstates and eigenvalues of the invariant operator $\hat{I}(t)$ may be found by an operator technique that is completely analogous to the method introduced by Dirac for diagonalizing the Hamiltonian of a constant-frequency harmonic oscillator”. In other words, from this point, the steps for finding the eigenstates and eigenvalues of $\hat{I}(t)$ are similar to those discussed in Sec. II and used to find the eigenstates and eigenvalues of $\hat{H}_0$. We start by defining the operators $\hat{a}(t)$ and $\hat{a}^\dagger(t)$ as [14]

$$\hat{a}(t) = \frac{1}{\sqrt{2\hbar}} \left\{ \frac{\hat{x}}{\rho(t)} + i[\rho(t)\hat{p} - m_0\dot{\rho}(t)\hat{x}] \right\}, \quad (23)$$

and

$$\hat{a}^\dagger(t) = \frac{1}{\sqrt{2\hbar}} \left\{ \frac{\hat{x}}{\rho(t)} - i[\rho(t)\hat{p} - m_0\dot{\rho}(t)\hat{x}] \right\}. \quad (24)$$

From these expressions, one can obtain that $\hat{a}(t)$ and $\hat{a}^\dagger(t)$ satisfy the commutation relation $[\hat{a}(t), \hat{a}^\dagger(t)] = 1$. By knowing this, the next step is to express the operator $\hat{I}(t)$ [Eq. (21)] in terms of $\hat{a}(t)$ and $\hat{a}^\dagger(t)$, which one obtains that

$$\hat{I}(t) = \hbar \left[\hat{a}^\dagger(t)\hat{a}(t) + \frac{1}{2} \right]. \quad (25)$$

In the sequence, we define the number operator $\hat{N}(t) = \hat{a}^\dagger(t)\hat{a}(t)$, whose eigenvalue and eigenstate are given by n and $|n; t\rangle$, respectively, so that one writes [14]

$$\hat{N}(t)|n; t\rangle = n|n; t\rangle, \quad (26)$$

with $n \in \mathbb{N}$. Thus, by means of Eqs. (9), (25) and (26), one sees that the eigenstate of the operator $\hat{I}(t)$ can be simply written as $|n; t\rangle$ [i.e., we can do the mapping $|\lambda, \zeta; t\rangle \rightarrow |n; t\rangle$], and its eigenvalues are given by $\lambda_n = \hbar(n + 1/2)$. Finally, defining the eigenstates of the invariant operator $\hat{I}(t)$ in position space, namely eigenfunctions, as $\Phi_n(x, t) = \langle x|n; t\rangle$, we have (see Appendix F for more details) [15, 16]:

$$\Phi_n(x, t) = \frac{1}{\sqrt{2^n n!}} \Phi_0(x, t) \mathcal{H}_n \left[\frac{x}{\hbar^{\frac{1}{2}} \rho(t)} \right], \quad (27)$$

where $\mathcal{H}_n$ are the Hermite polynomials of order n , and

$$\Phi_0(x, t) = \left[\frac{1}{\pi \hbar \rho(t)^2} \right]^{\frac{1}{4}} \exp \left\{ \frac{im_0}{2\hbar} \left[\frac{\dot{\rho}(t)}{\rho(t)} + \frac{i}{m_0 \rho(t)^2} \right] x^2 \right\}. \quad (28)$$

Note that, like the operator $\hat{I}(t)$, the eigenfunctions $\Phi_n(x, t)$ are completely determined if we know the parameter $\rho(t)$. Moreover, the eigenfunctions $\Phi_n(x, t)$ clearly form a complete orthonormal set (as required), since the Hermite polynomials constitute a basis of Hilbert space [104]. The last step is to find the general solution of the Schrödinger equation [Eq. (16)]. For this, we have to calculate the phase functions $\alpha_{\lambda, \zeta}(t)$ for this case by means of Eq. (15).

3. The phase functions

For the TDHO, we can do the mapping $\alpha_{\lambda, \zeta}(t) \rightarrow \alpha_n(t)$ in Eq. (15), so that it can be written as

$$\frac{d\alpha_n(t)}{dt} = \langle n; t | \left[i \frac{\partial}{\partial t} - \frac{1}{\hbar} \hat{H}(t) \right] | n; t \rangle. \quad (29)$$

Let us manipulate each term on the right side of the equation separately. For the first term, $i \langle n; t | \frac{\partial}{\partial t} | n; t \rangle$, we find (see Appendix G for more details)

$$i \langle n; t | \frac{\partial}{\partial t} | n; t \rangle = -\frac{1}{2} \left(n + \frac{1}{2} \right) m_0 [\rho(t)\ddot{\rho}(t) - \dot{\rho}(t)^2], \quad (30)$$

and, for the second term on the right side of Eq. (29), $(1/\hbar)\langle n; t | \hat{H}(t) | n; t \rangle$, we obtain

$$\frac{1}{\hbar} \langle n; t | \hat{H}(t) | n; t \rangle = \frac{1}{2m_0} \left(n + \frac{1}{2} \right) [\rho(t)^{-2} + m_0^2 \dot{\rho}(t)^2 + m_0^2 \omega(t)^2 \rho(t)^2]. \quad (31)$$

Thus, substituting Eqs. (30) and (31) into Eq. (29), and using Eq. (22), one obtains that the phase function $\alpha_n(t)$ is given by [14–16, 60]:

$$\alpha_n(t) = - \left(n + \frac{1}{2} \right) \int_0^t \frac{dt'}{m_0 \rho(t')^2}. \quad (32)$$

4. The general solution of the Schrödinger equation

For the TDHO, Eq. (16) is rewritten as $|\psi(t)\rangle = \sum_{n=0}^{\infty} C_n \exp[i\alpha_n(t)] |n; t\rangle$. By applying $\langle x|$ in both sides of this equation, one obtains $\Psi(x, t) = \sum_{n=0}^{\infty} C_n \Psi_n(x, t)$, where $\Psi(x, t) = \langle x | \psi(t) \rangle$ and $\Psi_n(x, t) = \exp[i\alpha_n(t)] \Phi_n(x, t)$. In this way, using Eqs. (27) and (32), one finally finds the wave function $\Psi_n(x, t)$, which is given by [15–17, 105]

$$\begin{aligned} \Psi_n(x, t) = & \frac{1}{\sqrt{2^n n! \pi^{\frac{1}{2}} \hbar^{\frac{1}{2}} \rho(t)}} \exp \left[-i \left(n + \frac{1}{2} \right) \int_0^t \frac{dt'}{m_0 \rho(t')^2} \right] \\ & \times \exp \left\{ \frac{im_0}{2\hbar} \left[\frac{\dot{\rho}(t)}{\rho(t)} + \frac{i}{m_0 \rho(t)^2} \right] x^2 \right\} \mathcal{H}_n \left[\frac{x}{\hbar^{\frac{1}{2}} \rho(t)} \right]. \end{aligned} \quad (33)$$

One can conclude that the initial problem is reduced to simply obtain the solution of the parameter $\rho(t)$ by means of Eq. (22), which, although is a nonlinear differential equation, has an exact solution. This is equivalent to solving a classical harmonic oscillator with time-dependent frequency (see Appendix E for more details), as Husimi concluded in 1953 [13].

5. Static case

For the particular case where the mass and the frequency of the oscillator are independent of time [$m(t) = m_0$ and $\omega(t) = \omega_0$], the parameter $\rho(t)$ takes the form [14, 16, 17]

$$\rho(t) = \rho_0 = \frac{1}{\sqrt{m_0 \omega_0}}. \quad (34)$$

In this case, it is immediate to see that the Hamiltonian $\hat{H}(t)$ falls back to the Hamiltonian $\hat{H}_0$ of a time-independent oscillator, and that the invariant operator becomes $\hat{I}(t) = \hat{I}_0 = \hat{H}_0/\omega_0$ [14], and the operators $\hat{a}(t)$ and $\hat{a}^\dagger(t)$ become, respectively, $\hat{a}_0$ and $\hat{a}_0^\dagger$ [Eqs. (A3) and (A4)]. Thus, the wave function $\Psi_n(x, t)$ recover the wave function $\Psi_n^{(0)}(x, t)$ related to the time-independent oscillator, given by Eq. (4), as expected. Similar to the previous section, by comparing the characteristics of a TDHO problem within the context of the LR method with the corresponding time-independent case, we can find correlations, which are summarized in Table II.

	Time-independent case	Time-dependent case
Schrödinger equation	$i\hbar \partial_t \psi(t)\rangle = \hat{H}_0 \psi(t)\rangle$	$i\hbar \partial_t \psi(t)\rangle = \hat{H}(t) \psi(t)\rangle$
Invariant operator	$\hat{H}_0$ given in Eq. (2)	$\hat{I}(t)$ given in Eq. (21)
Eigenvalue equation	$\hat{H}_0 \Phi_n^{(0)}(x, t) = \hbar \omega_0 (n + 1/2) \Phi_n^{(0)}(x, t)$	$\hat{I}(t) \Phi_n(x, t) = \hbar (n + 1/2) \Phi_n(x, t)$
Eigenfunction	$\Phi_n^{(0)}(x, t)$ given by Eq. (A15)	$\Phi_n(x, t)$ given by Eq. (27)
General solution	$\Psi^{(0)}(x, t) = \sum_{n=0}^{\infty} C_n^{(0)} \exp[i\alpha_n^{(0)}(t)] \Phi_n^{(0)}(x, t)$	$\Psi(x, t) = \sum_{n=0}^{\infty} C_n \exp[i\alpha_n(t)] \Phi_n(x, t)$
Phase function	$\alpha_n^{(0)}(t) = -(n + 1/2) \omega_0 t$	$\alpha_n(t) = -(n + 1/2) \int_0^t dt' / [m_0 \rho(t')^2]$

Table II. Comparison between the time-independent and time-dependent quantum harmonic oscillators.

B. The time-dependent quantum harmonic oscillator and the generation of squeezed states

In quantum physics, it is not possible to know the exact position and momentum of a particle at the same time, due to Heisenberg's uncertainty principle. This principle establishes a fundamental limit to the precision with which these two quantities can be measured simultaneously, which is given by $\sigma_{\hat{x}}\sigma_{\hat{p}} \geq \hbar/2$ (for the ground state) [7, 9], where $\sigma_{\hat{x}}$ and $\sigma_{\hat{p}}$ are the standard deviation associated with the position and momentum measurements, respectively. In the case of the quantum harmonic oscillator, when its frequency is modulated in a nonadiabatic way (that is, depends explicitly on the time), the quantum state, which was initially in equilibrium, is transformed into a so-called "squeezed" state, as pointed out in Refs. [15, 16, 24, 49, 76, 106]. In this sense, TDHOs provide a natural setting for studying these states. The squeezed states occur when the variance of an observable (for example, position) becomes smaller than its value for a coherent state, leading to an increase in the variance of the complementary observable (for example, momentum) in such a way that the Heisenberg's uncertainty principle is still satisfied [76, 107]. These states are fundamental for applications in quantum metrology, quantum sensing and quantum computing [52, 53].

To illustrate how to quantify the squeezing in an oscillator, it can be defined the parameters that characterize when the oscillator states are squeezed, namely the squeezing parameter $r(t)$ and the squeezing phase $\varphi(t)$, given respectively by (for more details, see Refs. [56, 78, 108])

$$r(t) = \cosh^{-1} \{ [\lambda(t)]^{\frac{1}{2}} \}, \quad (35)$$

where

$$\lambda(t) = \frac{m_0^2 \dot{\rho}(t)^2 + \rho(t)^{-2} + m_0^2 \omega(t)^2 \rho(t)^2 + 2m_0 \omega(t)}{4m_0 \omega(t)}, \quad (36)$$

and

$$\varphi(t) = \cos^{-1} \left\{ \frac{1 + m_0 \omega(t) \rho(t)^2 - 2 \cosh^2[r(t)]}{2 \sinh[r(t)] \cosh[r(t)]} \right\}. \quad (37)$$

Note that, when the oscillator is time-independent, the parameter $\rho(t)$ is given by Eq. (34) and when substituting it into Eqs. (35)-(37), one obtains that the squeezing parameter is null and the squeezing phase is undefined. Thus, for a time-independent harmonic oscillator, there are no squeezed states, as expected.

Taking into account the wave function $\Psi_n(x, t)$ [Eq. (33)], we can define the expected value of a given observable $\hat{O}$ in the state $\Psi_n(x, t)$ for any $t \geq 0$ as [7, 9]

$$\langle \hat{O} \rangle(n, t) = \int_{-\infty}^{+\infty} dx \Psi_n^*(x, t) \hat{O} \Psi_n(x, t). \quad (38)$$

Using this equation and the definition of the variance associated with the operator $\hat{O}$, namely $\sigma_{\hat{O}}^2(n, t) = [\langle \hat{O}^2 \rangle(n, t)] - [\langle \hat{O} \rangle(n, t)]^2$, we have, for the position and momentum operators, that [56]

$$\sigma_{\hat{x}}^2(n, t) = \{ \cosh^2[r(t)] + 2 \sinh[r(t)] \cosh[r(t)] \cos[\varphi(t)] + \sinh^2[r(t)] \} \left(n + \frac{1}{2} \right) \frac{\hbar}{m_0 \omega(t)}, \quad (39)$$

$$\sigma_{\hat{p}}^2(n, t) = \{ \cosh^2[r(t)] - 2 \sinh[r(t)] \cosh[r(t)] \cos[\varphi(t)] + \sinh^2[r(t)] \} \left(n + \frac{1}{2} \right) \hbar m_0 \omega(t). \quad (40)$$

As discussed, in the time-independent case $[\omega(t) = \omega_0]$ we have $r(t) = 0$, so that Eqs. (39) and (40) recover the usual variances of a time-independent harmonic oscillator [7]. If we choose, for example and simplicity, $\varphi(t) = 0$ in Eqs. (39) and (40), we obtain

$$\sigma_{\hat{x}}^2(n, t)|_{\varphi(t)=0} = \exp[2r(t)] \left(n + \frac{1}{2} \right) \frac{\hbar}{m_0 \omega(t)}, \quad \sigma_{\hat{p}}^2(n, t)|_{\varphi(t)=0} = \exp[-2r(t)] \left(n + \frac{1}{2} \right) \hbar m_0 \omega(t). \quad (41)$$

From Eqs. (41), one can see that while the ratio $\sigma_{\hat{x}}^2(n, t)|_{\varphi(t)=0} / \left[(n + 1/2) \frac{\hbar}{m_0 \omega(t)} \right]$ (involving the variance in the position) enhances exponentially with time, the ratio $\sigma_{\hat{p}}^2(n, t)|_{\varphi(t)=0} / [(n + 1/2) \hbar m_0 \omega(t)]$ (involving the variance in the momentum) decreases (or is squeezed) exponentially.

Furthermore, it is known that the system subjected to a TDHO potential can also make transitions between different states. In this sense, the probability $P(t)_{0 \rightarrow \nu}$ of the oscillator making a transition from its ground state to an excited state ν is defined by [76, 107, 109]

$$P(t)_{0 \rightarrow \nu} = \frac{\nu! \tanh^\nu[r(t)]}{2^\nu [(\nu/2)!]^2 \cosh[r(t)]}, \quad (42)$$

where $\nu = 0, 2, 4, \dots$. When we consider $\nu = 0$ in this equation, one has the probability of persistence in the fundamental state, $P_p(t) = 1/\cosh[r(t)]$. With this, one can also obtain the probability of excitation, which is given by $P_e(t) = 1 - P_p(t)$ [76]. We will apply these concepts in the following sections.

Now, let us discuss an analogy between a quantum and a classical TDHO. More specifically, the corresponding in the classical case of the generation of the squeezed states in the quantum. To this end, let us consider the scenario of the child playing on a swing in a park, as mentioned in Sec. I. If the child bends the knees and then extends the legs at the right time (see Fig. 1 of Ref. [11]), energy is transferred to the system child + swing (through the modulation of the center of mass). That situation create a pendulum with effective variable length over time. Mathematically, for motions with small amplitudes, this system is represented by a classical harmonic oscillator with an effective time-dependent oscillation frequency, which satisfies the equation [1]

$$\ddot{\Theta}(t) + \Omega(t)^2 \Theta(t) = 0, \quad (43)$$

where $\Theta(t)$ is the angular displacement, and the function $\Omega(t)$ is the oscillation frequency, which is periodic in time [$\Omega(t+T) = \Omega(t)$, where T is the oscillation period]. When the swing frequency is modulated as twice the natural oscillation frequency $\Omega_0 \equiv \Omega(t)|_{t=0}$, for example [1],

$$\Omega(t)^2 = \Omega_0^2 \{1 + h \cos[(2\Omega_0 + \epsilon)t]\}, \quad h \ll 1, \quad \epsilon \ll \Omega_0, \quad (44)$$

we have the so-called parametric resonance (or parametric amplification [11]), such that $\Theta(t)$ assumes the form [1]

$$\Theta(t) = \Theta_0^{(1)} \exp(st) \cos\left[\left(\Omega_0 + \frac{\epsilon}{2}\right)t\right] + \Theta_0^{(2)} \exp(-st) \sin\left[\left(\Omega_0 + \frac{\epsilon}{2}\right)t\right], \quad (45)$$

in which $\Theta_0^{(1)}$ and $\Theta_0^{(2)}$ are constants (particularly, $\Theta_0^{(1)}$ is the initial amplitude of the motion), $s = \frac{1}{2}\sqrt{\left(\frac{h\Omega_0}{2}\right)^2 - \epsilon^2}$, and $-\frac{h\Omega_0}{2} < \epsilon < \frac{h\Omega_0}{2}$ (for ϵ to be a real quantity). In this way, one of the motion components [the first term on the right hand side of Eq. (45)] increases exponentially (the amplitude increases), while the other decreases exponentially (or is “squeezed”). Note that this is analogous to what happens in the quantum case shown in Eqs. (41), where the ratio related to the variance in the position increases exponentially with time, whereas the ratio related to the variance in momentum decreases (or is squeezed) exponentially. For a more detailed discussion on this analogy, see Ref. [11].

C. Application to an undergraduate and postgraduate problem

In this section, by means of the results presented in previous sections, we solve the problem 2.14 of Ref. [9] (or the problem 11.19 of Ref. [12]), which reads as follows: “A particle is in the ground state of the harmonic oscillator with classical frequency ω_0 , when suddenly the spring constant quadruples, so $\omega_1 = 2\omega_0$, without initially changing the wave function. What is the probability that a measurement of the energy would still return the value $\hbar\omega_0/2$? What is the probability of getting $\hbar\omega_0$?”

Let us start discussing this problem considering a general value for ω_1 . In a general way, this problem can be modeled as a TDHO with a time-dependent frequency defined by

$$\omega(t) = \begin{cases} \omega_0, & t < 0, \\ \omega_1, & t \geq 0. \end{cases} \quad (46)$$

In this case, the energy spectrum in the interval $t \geq 0$ simply becomes $E'_n = \hbar\omega_1(n + 1/2)$. Besides this, as discussed, quantum harmonic oscillators with time-dependent parameters describe squeezed states. To calculate the squeezing parameter $r(t)$, we first have to calculate the parameter $\rho(t)$, which is calculated from the Ermakov-Pinney equation [Eq. (22)]. One can find that the solution of the Ermakov-Pinney equation for the interval $t \geq 0$, and subjected to the initial conditions $\rho(t)|_{t=0} = \rho_0$ and $\dot{\rho}(t)|_{t=0} = 0$, is given by [55]

$$\rho(t) = \sqrt{\frac{\omega_0 \sin^2(\omega_1 t)}{m_0 \omega_1^2} + \frac{\cos^2(\omega_1 t)}{m_0 \omega_0}}. \quad (47)$$

Substituting this result into Eqs. (35) and (36), we obtain

$$r(t) = \cosh^{-1} \left(\frac{\omega_0 + \omega_1}{2\sqrt{\omega_0 \omega_1}} \right). \quad (48)$$

Moreover, since the system is initially in the ground state, it is also interesting to calculate the probability of its persistence in the ground state. For this purpose, substituting Eq. (48) in Eq. (42) with $\nu = 0$, one obtains

$$P_p(t) = \frac{2\sqrt{\omega_0 \omega_1}}{\omega_0 + \omega_1}. \quad (49)$$

Note that in both limits $\omega_1/\omega_0 \rightarrow 0$ and $\omega_1/\omega_0 \rightarrow \infty$, we have $P_p(t) \rightarrow 0$, which means that after the system changes its frequency in these limit situations, it will also be excited. The behavior of $P_p(t)$ given by Eq. (49) is illustrated in Fig. 1.

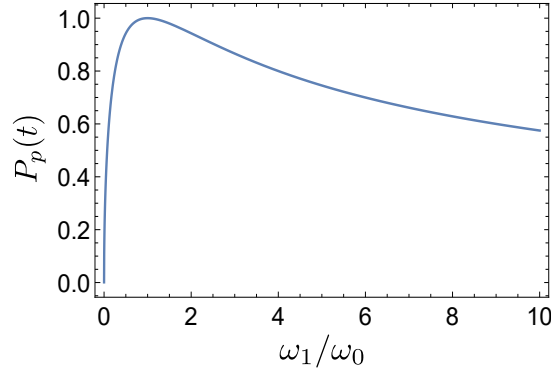


Figure 1. Behavior of $P_p(t)$ given by Eq. (49), as a function of ω_1/ω_0 .

For the problem 2.14 of Ref. [9] (or the problem 11.19 of Ref. [12]), we have $\omega_1 = 2\omega_0$. In this case, the energy spectrum in the interval $t \geq 0$ becomes $E'_n = 2\hbar\omega_0 (n + 1/2)$. Note that, in this interval, the energy associated to the ground state ($n = 0$) is $E'_0 = \hbar\omega_0$. This means that, when $t \geq 0$, the probability of a measurement of the energy resulting in any value less than E'_0 is zero and, thus, such probability for $\hbar\omega_0/2$ is zero. On the other hand, the probability of getting $\hbar\omega_0$ may not be zero and is simply given by Eq. (49) with $\omega_1 = 2\omega_0$ (since $\hbar\omega_0$ is the energy corresponding to the ground state for $t \geq 0$) which results in $P_p(t) = \frac{2\sqrt{2}}{3} \cong 0.943$. This result is the same as that obtained from the elementary quantum mechanics methods discussed in undergraduate and postgraduate courses (see Refs. [9, 12]).

It is worth noting that alternative methods can be used to solve the same problem. For example, in Refs. [49, 51, 76, 110], instead of deriving the wave functions $\Psi_n(x, t)$ after the frequency change (as done here, as well as in Refs. [9, 12]), in those articles, using the Heisenberg picture, through the Bogoliubov transformations relating annihilation and creation operators before and after the frequency changes, it becomes possible to write the squeezing parameter $r(t)$ in terms of these coefficients, and then, by means of Eq. (42), one can obtain the same result as found in Eq. (49).

Next, we analyze a model where the usual methods taught at undergraduate and postgraduate level cannot be applied directly.

D. Quantum dynamics of a particle in a Paul trap

Paul traps, also known as radio-frequency traps, are devices used to capture and manipulate ions using oscillating electric fields [27, 34, 111]. More specifically, these fields create an oscillating quadrupole potential that generates an average force capable of confining particles in all directions. Developed by German physicist Wolfgang Paul (who received the Nobel Prize for physics in 1989 for this innovation), they have applications in different areas of physics as, for instance, mass spectroscopy, cooling processes, quantum computing, among others (for more details, see Refs. [27, 34, 111]).

For a classic mechanical analogy (which can help visualize how the device functions), consider a ball in a potential that has the form of a “horse saddle”, in which the ball is fixed at the center of the saddle (see Ref. [112], where Fig. 1 provides an illustration of this model). If the saddle is static, the ball does not stay at the center for a long time. Indeed, it rolls downwards in the unstable direction. In the other direction, the ball is in stable equilibrium because the upward curvature prevents it from falling. In other words, the static saddle is a stable potential in one direction and unstable in another. On the other hand, if the saddle rotates rapidly around its vertical axis (like a carousel), the ball no longer falls: it remains “trapped” at the center of the saddle (exhibiting only small oscillations around this point), since the constant rotation changes the directions of stability and instability so quickly that the ball does not have time to escape. For a clarifying discussion, see Ref. [112] and references therein.

In this sense, due to the importance of this device for the modern physics, it could be very enriching to make a basic discussion on this topic in undergraduate and postgraduate quantum mechanics courses. However, the tools usually taught in these courses are not enough to deal with this problem, since the dynamics of a particle in a Paul trap is effectively modeled, for each of the three Cartesian coordinates of the center of mass of the trapped particle, by a TDHO with time-dependent frequency of the form [27, 102, 111]

$$\omega(t) = \frac{\omega_0}{\sqrt{\beta + \gamma}} \sqrt{\beta + \gamma \cos\left(\frac{2\pi t}{\tau}\right)}, \quad (50)$$

where τ is the oscillation period [thus, $\omega(t + \tau) = \omega(t)$], and β and γ are numerical factors such that $\beta > \gamma$. By means of the LR method, this problem can be solved exactly and, thus, it can be interesting to make a brief analyzes of this case (something similar has already been done, for example, in Ref. [102]).

Using Eqs. (50) and (H2) (see Appendix H) in Eqs. (35) and (36), we obtain the squeezing parameter $r(t)$ for this model, whose behavior is shown in Fig. 2. Note the periodic oscillatory behavior of the squeezing parameter $r(t)$ [102], which occurs

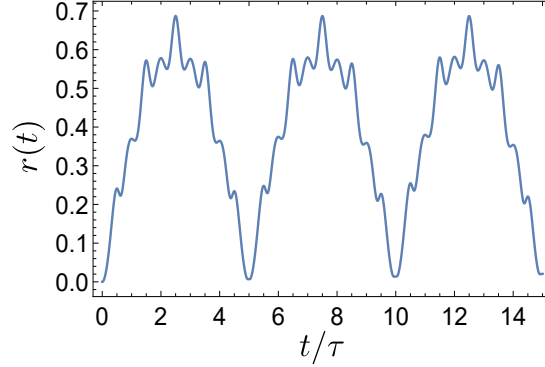


Figure 2. Behavior of the squeezing parameter $r(t)$ associated with the model defined by Eq. (50), as a function of t/τ . In this figure we consider, in arbitrary units, $\beta = 1$, $\gamma = 1/2$ and $\omega_0\tau = 3$.

naturally due to the form of the frequency $\omega(t)$ in Eq. (50). This also influences the behavior of all the other associated quantities [see Eqs. (39)-(42)], as can be seen in Fig. 3 for the variances of position and momentum, and in Fig. 4 for the probability of persistence in the ground state $P_p(t)$ and the probability of excitation $P_e(t)$. Note that when the values of

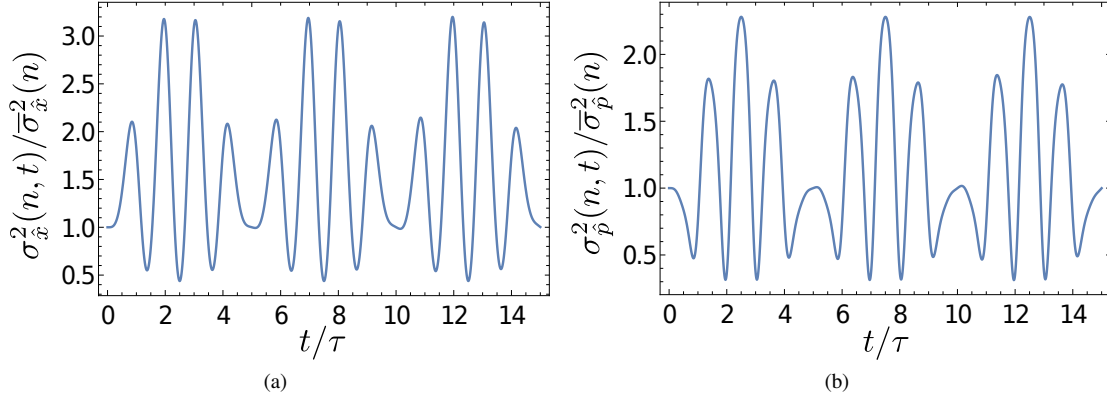


Figure 3. Behavior of the variances (a) $\sigma_x^2(n, t)/\bar{\sigma}_x^2(n)$ and (b) $\sigma_p^2(n, t)/\bar{\sigma}_p^2(n)$, with $\bar{\sigma}_x^2(n) = \frac{\hbar(n+1/2)}{m_0\omega_0}$ and $\bar{\sigma}_p^2(n) = \hbar m_0\omega_0(n + 1/2)$, as a function of t/τ . In these figures we consider, in arbitrary units, $\beta = 1$, $\gamma = 1/2$ and $\omega_0\tau = 3$.

$\sigma_x^2(n, t)$ are increasing in Fig. 3(a), the corresponding values of $\sigma_p^2(n, t)$ in Fig. 3(b) are decreasing, and vice-versa, being in agreement with Heisenberg's uncertainty principle. In a similar way, when the values of $P_p(t)$ are increasing in Fig. 4(a), the corresponding values of $P_e(t)$ in Fig. 4(b) are decreasing, and vice-versa, since $P_p(t) + P_e(t) = 1$. In addition, Figs. 2 and 4(b) show that $P_e(t)$ assumes the maximum values in the same times as $r(t)$ takes on its maximum values. This occurs because $P_e(t) = 1 - 1/\cosh[r(t)]$, i.e., the higher the value of $r(t)$, $P_e(t)$ approaches to 1.

V. FINAL REMARKS

In Sec. II, we briefly reviewed the algebraic method to obtain the solution of the Schrödinger equation for the problem of a time-independent quantum harmonic oscillator, which is widely studied in quantum mechanics courses [7–9]. This section prepares the reader for the subsequent sections, since the application of the Lewis-Riesenfeld method to a time-dependent harmonic oscillator prescribes a sequence of steps similar to those already used in the time-independent case (and which were summarized in the end of Sec. II).

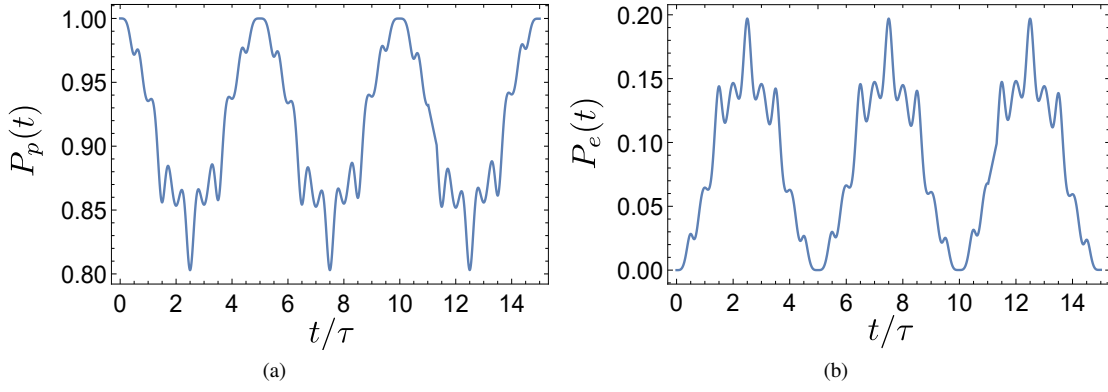


Figure 4. Behavior of the probabilities (a) $P_p(t)$ and (b) $P_e(t)$, as a function of t/τ . In these figures we consider, in arbitrary units, $\beta = 1$, $\gamma = 1/2$ and $\omega_0\tau = 3$.

In Sec. III, we discussed the general formulation of the Lewis-Riesenfeld method [14, 60, 61], step-by-step and with a didactic approach, and we presented a summary of the method, organized into four steps. We highlighted that the usual way to approach the problem of the time-independent quantum harmonic oscillator (looking for eigenfunctions of the time-independent Hamiltonian operator) is a particular case of the Lewis-Riesenfeld approach (looking for eigenfunctions of the Lewis' invariant operator). We also presented Table I, where we summarized a comparison between the characteristics of a general approach to obtain the solution of a time-independent case and the correspondent ones of a time-dependent situation.

In Sec. IV A, we provided a didactic application of the Lewis-Riesenfeld method, obtaining the wave function for the case of a time-dependent harmonic oscillator with a generic time-dependent frequency. We highlighted that the steps for finding the eigenstates and eigenvalues of the Lewis' invariant operator for this system (Lewis' invariant operator is written in terms of ladder operators) are similar to those of the usual algebraic method used to find the eigenstates and eigenvalues of the Hamiltonian of a time-independent quantum harmonic oscillator (the time-invariant Hamiltonian operator is written in terms of ladder operators). Closing this section, we presented Table II, where we summarized a comparison between the characteristics of the time-independent oscillator addressed via algebraic approach (discussed in Sec. II) and the correspondent ones of a time-dependent oscillator addressed via Lewis-Riesenfeld method.

In Sec. IV B, we briefly discussed the relation between the quantum time-dependent harmonic oscillator and the generation of squeezed states, using, in Sec. IV C, these concepts to calculate the transition probability for a time-dependent harmonic oscillator experiencing a sudden jump in its frequency [9, 12]. In Sec. IV D, we also used all these tools to briefly study the dynamics of a quantum particle in a Paul trap [27, 102]. These calculations performed in Secs. IV C and IV D can be used as didactic examples of applications of the Lewis-Riesenfeld method, which exactly solves the Schrödinger equation with explicitly time-dependent Hamiltonians, in quantum mechanics courses.

The presented discussion aims to contribute to the effort already begun in articles in physics teaching journals [10, 18, 57–59] to improve the discussions on time-dependent quantum harmonic oscillators in the context of undergraduate courses.

DECLARATION OF COMPETING INTEREST

The authors declare no conflict of interest.

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Appendix A: Time-independent harmonic oscillator: a brief review

First, we write the function $\Psi_n^{(0)}(x, t)$ in Eq. (3) as

$$\Psi_n^{(0)}(x, t) = \Phi_n^{(0)}(x) \exp\left(-\frac{iE_n t}{\hbar}\right), \quad (\text{A1})$$